3.1.2 Linear motion

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)** the equations of motion for constant acceleration in a straight line, including motion of bodies falling in a uniform gravitational field without air resistance $v = u + at \quad s = \frac{1}{2}(u + v)t$	*M2.2, M2.4, M3.3*

(a)	**(ii)** techniques and procedures used to investigate the motion and collisions of objects	**PAG1** Apparatus may include trolleys, air-track gliders, ticker timers, light gates, data-loggers and video techniques.
(b)	**(i)** acceleration *g* of free fall	
	(ii) techniques and procedures used to determine the acceleration of free fall using trapdoor and electromagnet arrangement or light gates and timer	**PAG1** HSW5 Determining *g* in the laboratory.
(c)	reaction time and thinking distance; braking distance and stopping distance for a vehicle.	